so the diffusion term is

$$\beta^{-1}\mathrm{Tr}\left(\boldsymbol{\Gamma}^{1/2}\boldsymbol{M}^{1/2}[\beta\boldsymbol{M}^{-1}\boldsymbol{p}\boldsymbol{p}^{T}\boldsymbol{M}^{-1}-\boldsymbol{M}^{-1}]\boldsymbol{M}^{1/2}\boldsymbol{\Gamma}^{1/2}\right)\bar{\rho}=-\mathrm{drift},$$

where we have used the cyclic property of the trace (and the symmetry of $\boldsymbol{\Gamma}$) to reduce the expression. The validity of the term involving $\boldsymbol{\xi}$ is easily verified by a similar argument.

Thus we know that the Gibbs distribution will be invariant. It remains to prove that the SDE (8.22)–(8.24) is ergodic. Certainly this would be expected in the case where a full Langevin dynamics contribution is present ($\boldsymbol{\Gamma}$ positive definite), but as we shall see next there are cases where noise only contacts the auxiliary degrees of freedom but the system is nonetheless ergodic.

We state here a useful observation that allows to simplify the commutator calculations in the case of an SDE of the following type:

$$\mathrm{d}\boldsymbol{z}=\boldsymbol{F}(\boldsymbol{z})\mathrm{d}t+\xi\boldsymbol{G}(\boldsymbol{z})\mathrm{d}t, \tag{8.25}$$

$$\mathrm{d}\xi=g(\boldsymbol{z})\mathrm{d}t-\gamma\xi\mathrm{d}t+\sigma\mathrm{dW}, \tag{8.26}$$

where $\boldsymbol{z}\in\mathbb{R}^m$, $\boldsymbol{F},\boldsymbol{G}:\mathbb{R}^m\to\mathbb{R}^m$.

Proposition 8.2 *If the Lie algebra generated by $\boldsymbol{F}$ and $\boldsymbol{G}$ in (8.25) spans $\mathbb{R}^m$, then the Hörmander condition holds for the system (8.25)–(8.26).*

Proof First observe that the premise holds if and only if the Lie algebra of the vectors in $\mathbb{R}^{m+1}$ defined by

$$\tilde{\boldsymbol{F}}=\begin{bmatrix}\boldsymbol{F}\\0\end{bmatrix},\qquad\tilde{\boldsymbol{G}}=\begin{bmatrix}\boldsymbol{G}\\0\end{bmatrix},$$

together with $\boldsymbol{e}_{m+1}$ spans $\mathbb{R}^{m+1}$. Thus the problem is to show that we can generate $\tilde{\boldsymbol{F}}$ and $\tilde{\boldsymbol{G}}$ from the vector fields that define (8.25)–(8.26), which are

$$\boldsymbol{b}_0=\begin{bmatrix}\boldsymbol{F}+\xi\boldsymbol{G}\\g-\gamma\xi\end{bmatrix},\qquad\boldsymbol{b}_1=\begin{bmatrix}\boldsymbol{0}\\\sigma\end{bmatrix}.$$

Computing the commutator gives

$$\boldsymbol{c}=[\boldsymbol{b}_0,\boldsymbol{b}_1]=\begin{bmatrix}-\sigma\boldsymbol{G}\\\sigma\gamma\end{bmatrix},$$

hence

$$\tilde{\boldsymbol{G}}=\frac{-1}{\sigma}\left[\boldsymbol{c}-\gamma\boldsymbol{b}_1\right].$$

This means that we have built one of our two vectors. The second follows as a linear combination of the others:

$$\tilde{\boldsymbol{F}} = \boldsymbol{b}_0 - \xi\tilde{\boldsymbol{G}} - (g - \gamma\xi)\sigma^{-1}\boldsymbol{b}_1.$$

□

The fact that the Hörmander condition holds is, as we know from Chap. 6, only part of the story. However based on known results for Langevin dynamics, we conjecture that ergodicity will hold if (i) the Hörmander condition holds, (ii) U is sufficiently smooth, and (iii) the configurational phase space is compact, e.g. periodic boundary conditions are employed. However we stress that each possible method will ultimately need to be carefully and systematically checked to verify the ergodic properties.

8.4.1 *Nosé-Hoover-Langevin: An Ergodic Weakly Coupled Thermostat*

The Nosé-Hoover-Langevin (NHL) method is based on a simple idea: replace the "chain" in the Nosé-Hoover Chain, whose sole purpose is to maintain a Gaussian distribution in the auxiliary variable, by a stochastic Langevin-type thermostat. The method was first proposed in [323]. The proof of ergodicity (more precisely the confirmation of the Hörmander condition), for a problem with harmonic internal interactions, was given in [226] and we roughly follow the treatment from this paper.

The Nosé-Hoover-Langevin equations are

$$\mathrm{d}\boldsymbol{q} = \boldsymbol{M}^{-1}\boldsymbol{p}\mathrm{d}t, \tag{8.27}$$

$$\mathrm{d}\boldsymbol{p} = -\nabla U(\boldsymbol{q})\mathrm{d}t - \xi\boldsymbol{p}\mathrm{d}t, \tag{8.28}$$

$$\mathrm{d}\xi = \mu^{-1}[\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d k_B T}]\mathrm{d}t - \gamma\xi\mathrm{d}t + \sqrt{2\mathrm{k_B T}\gamma\mu^{-1}}\mathrm{dW}, \tag{8.29}$$

where $(\boldsymbol{q},\boldsymbol{p}) \in \mathcal{D}$ with $\xi \in \mathbb{R}$. Let us emphasize that the principal features of this thermostat: (1) it is essentially Nosé-Hoover plus a perturbation, (2) it incorporates a single Wiener process, (3) it has two free parameters γ and μ.

Let us first demonstrate that the Nosé-Hoover-Langevin method, is, in fact, a thermostat. We observe that we have already shown that the Nosé-Hoover method preserves the extended canonical distribution $\rho_\beta \exp(-\beta\mu\xi^2/2)$. Therefore, by the additivity property, all that is needed is to show that the additional part

$$\mathrm{d}\xi = -\gamma\xi\mathrm{d}t + \sqrt{2\mathrm{k_B T}\gamma\mu^{-1}}\mathrm{dW}$$